Enterprise LAN Implementation with Inter-VLAN Routing

Computer Communication Networks (CSE248)

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Group Members

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1 Selected Network Type

The selected network type for this project is **Enterprise LAN with VLANs**. This architecture is chosen to simulate a structured campus or Educational enterprise where different types of internet usage exist and operate within same physical infrastructure but while remaining logically separated.

The use of Virtual Local Area Networks (VLANs) allows the segmentation of broadcast domains, improving both network performance and security. Inter-VLAN communication is achieved using the Router-on-a-Stick (RoAS) approach, where a single router interface is subdivided into multiple logical interfaces.

Our network design is an educational LAN network, where VLANs are classified according to their purpose. The VLAN10 (LABs) is used for a network of deivces in labs, which may have security restrictions like intalling or running packet sniffing tools like wireshark. The VLAN20 (Classroom internet) is used for wifi network in the classrooms. The VLAN30 (Operational) is used for network of devices which are used for operational purposes and doesn't needs to be on same network as classroom internet like on campus security cameras and printers.

2 Network Topology

2.1 Physical Topology

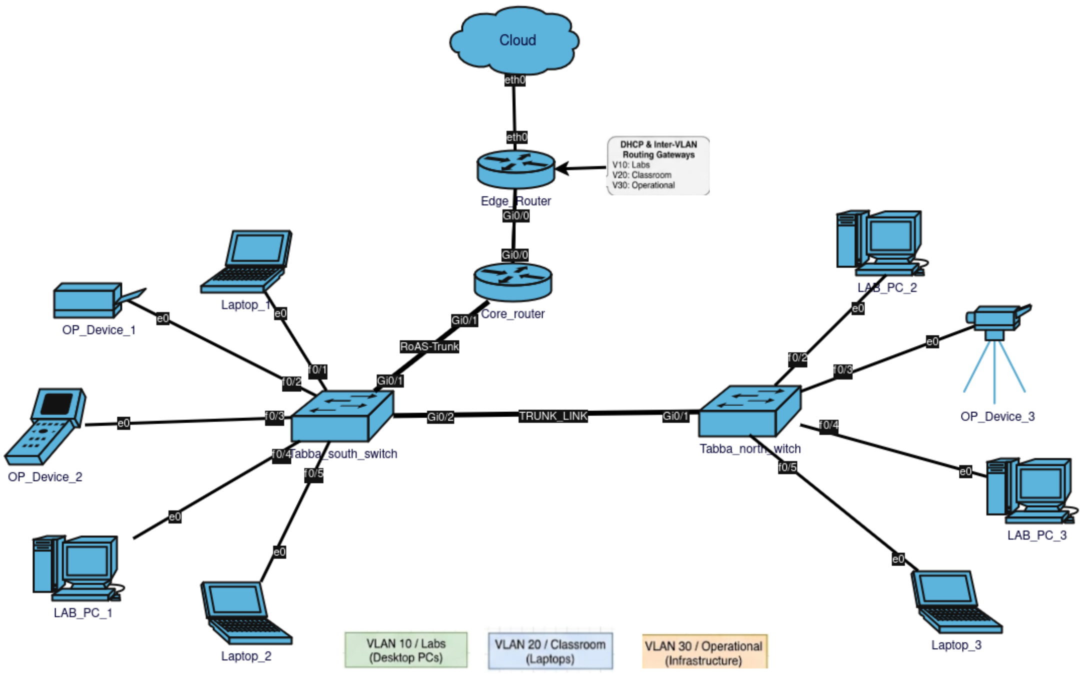

Figure 1: Physical Topology of the Network

The physical topology above consists of two routers and two switches interconnected to form a hierarchical network structure.The End devices are connected to access switches and the switches are connected with the core router with trunk links.The edge router provides connectivity to the external network.

2.2 Logical Topology

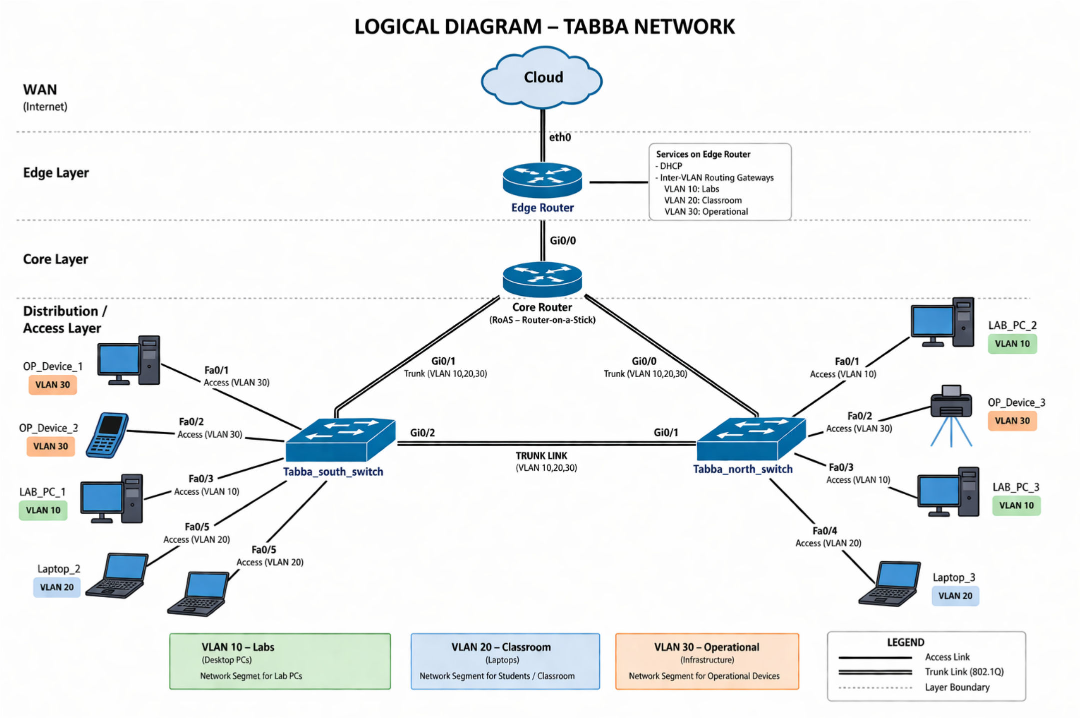

Figure 2: Logical Topology with VLAN Segmentation

The logical topology represents VLAN segmentation and IP addressing within the network.The Virtual Local Area Networks are configured to seperate different types of devices on the basis of usage : Labs, Classroom internet , network for Operational devices.Routing between these VLANs is handled centrally by the core router.

3 IP Addressing Plan

3.1 Subnet Allocation

Network Segment	VLAN	Network	Subnet Mask	Usable Host Range	Gateway
Labs	10	192.168.10.0	255.255.255.0 (/24)	192.168.10.1 – 192.168.10.254	192.168.10.1
Classroom	20	192.168.20.0	255.255.255.0 (/24)	192.168.20.1 – 192.168.20.254	192.168.20.1
Operational	30	192.168.30.0	255.255.255.0 (/24)	192.168.30.1 – 192.168.30.254	192.168.30.1
core-edge router link	–	10.1.1.0	255.255.255.252 (/30)	10.1.1.1 – 10.1.1.2	–
Public Link	–	203.0.113.0	255.255.255.252 (/30)	203.0.113.1 – 203.0.113.2	–

The addressing scheme follows a structured private IP allocation based on RFC 1918 standards. Each VLAN is assigned a dedicated /24 subnet to ensure logical segmentation, efficient broadcast management, and scalability for future expansion.

A /30 subnet is used for both the core router- edge router link and the public link, as these are point-to-point connections between two devices. This minimizes IP address wastage while maintaining efficient communication between the edge and core routers.

3.2 Router Interface Addressing

Device	Interface	IP Address	Purpose
Edge Router	eth0	203.0.113.2	External Link
Edge Router	Gi0/0	10.1.1.1	Backbone Link
Core Router	Gi0/0	10.1.1.2	Backbone Link
Core Router	Gi0/1.10	192.168.10.1	VLAN 10 Gateway
Core Router	Gi0/1.20	192.168.20.1	VLAN 20 Gateway
Core Router	Gi0/1.30	192.168.30.1	VLAN 30 Gateway

3.3 End Device Addressing

Device	VLAN	IP Address	Gateway
LAB_PC_1	10	192.168.10.10	192.168.10.1
LAB_PC_2	10	192.168.10.11	192.168.10.1
LAB_PC_3	10	192.168.10.12	192.168.10.1
Laptop_1	20	192.168.20.10	192.168.20.1
Laptop_2	20	192.168.20.11	192.168.20.1
Laptop_3	20	192.168.20.12	192.168.20.1
OP_Device_1	30	192.168.30.10	192.168.30.1
OP_Device_2	30	192.168.30.11	192.168.30.1
OP_Device_3	30	192.168.30.12	192.168.30.1

4 Services to be Implemented

The following network services are planned for implementation:

- **Inter-VLAN Routing:** Achieved using Router-on-a-Stick configuration on the core router.
- **DHCP:** Automatic IP address assignment for all VLANs.
- **NAT (PAT):** Translation of private IP addresses to a public IP for external communication.
- **DNS:** Domain name resolution using a public DNS server.
- **Static Routing:** Default route configured for external traffic forwarding.

5 Division of Work

Member	**Phase 1**	**Phase 2**
Abdulah Siraj	GNS3 setup, Linux setup on personal device, topology design	Router configuration, NAT, testing
Muhammad Yahya	IP addressing, documentation, Linux setup of personal device	VLAN config, DHCP, final report

6 Acknowledgment of Tools

While making this report limited assistance from a large Language Model tool was taken for improving formating and structuring draft .docx file into LaTeX format for PDF generation. All network design decisions, including topology creation, IP addressing scheme, and configuration planning, were independently developed and verified.

7 References

1. Rekhter, Y., et al., "Address Allocation for Private Internets (RFC 1918),"
2. Droms, R., "Dynamic Host Configuration Protocol (RFC 2131),"
3. Cisco Systems Documentation, "Inter-VLAN Routing and 802.1Q Trunking"